

OpenStax Calculus Volume 2

Section 5.4 Exercises: Comparison Tests

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Use the comparison test to determine whether the following series converge.

1. $\sum_{n=1}^{\infty} a_n$ where $a_n = \frac{2}{n(n+1)}$
2. $\sum_{n=1}^{\infty} a_n$ where $a_n = \frac{1}{n(n+1/2)}$
3. $\sum_{n=1}^{\infty} \frac{1}{2(n+1)}$
4. $\sum_{n=2}^{\infty} \frac{1}{2(n-1)}$
5. $\sum_{n=2}^{\infty} \frac{1}{(n \ln n)^2}$
6. $\sum_{n=1}^{\infty} \frac{n!}{(n+2)!}$
7. $\sum_{n=1}^{\infty} \frac{1}{n!}$
8. $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n^2}$
9. $\sum_{n=1}^{\infty} \frac{\sin^2 n}{n^2}$
10. $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{(\sqrt{n})^3}$
11. $\sum_{n=1}^{\infty} \frac{n^{1.2}-1}{n^{2.3}+1}$
12. $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{n}$
13. $\sum_{n=1}^{\infty} \frac{\sqrt[4]{n}}{\sqrt[3]{n^4+n^2}}$

Use the limit comparison test to determine whether each of the following series converges or diverges.

14. $\sum_{n=1}^{\infty} \left(\frac{\ln n}{n}\right)^2$
15. $\sum_{n=1}^{\infty} \left(\frac{\ln n}{n^{0.6}}\right)^2$
16. $\sum_{n=1}^{\infty} \frac{\ln(1+\frac{1}{n})}{n}$
17. $\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n^2}\right)$
18. $\sum_{n=1}^{\infty} \frac{1}{4^n-3^n}$

19. $\sum_{n=1}^{\infty} \frac{1}{n^2 - n \sin n}$
20. $\sum_{n=1}^{\infty} \frac{1}{e^{(1.1)^n} - 3^n}$
21. $\sum_{n=1}^{\infty} \frac{1}{e^{(1.01)^n} - 3^n}$
22. $\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}$
23. $\sum_{n=1}^{\infty} \frac{1}{2^{1+1/n} n^{1+1/n}}$
24. $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \sin \left(\frac{1}{n} \right) \right)$
25. $\sum_{n=1}^{\infty} \left(1 - \cos \left(\frac{1}{n} \right) \right)$
26. $\sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{\pi}{2} - \arctan n \right)$
27. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{n} \right)^n$ (Hint: $(1 - \frac{1}{n})^n \rightarrow 1/e$.)
28. $\sum_{n=1}^{\infty} (1 - e^{-1/n})$ (Hint: $1/e \approx (1 - 1/n)^n$, so $1 - e^{-1/n} \approx 1/n$.)
29. Does $\sum_{n=2}^{\infty} \frac{1}{(\ln n)^p}$ converge if p is large enough? If so, for which p ?
30. Does $\sum_{n=1}^{\infty} \left(\frac{\ln n}{n} \right)^p$ converge if p is large enough? If so, for which p ?
31. For which p does the series $\sum_{n=1}^{\infty} 2^{pn}/3^n$ converge?
32. For which $p > 0$ does the series $\sum_{n=1}^{\infty} \frac{n^p}{2^n}$ converge?
33. For which $r > 0$ does the series $\sum_{n=1}^{\infty} \frac{r^{n^2}}{2^n}$ converge?
34. For which $r > 0$ does the series $\sum_{n=1}^{\infty} \frac{2^n}{r^{n^2}}$ converge?
35. Find all values of p and q such that $\sum_{n=1}^{\infty} \frac{n^p}{(n!)^q}$ converges.
36. Does $\sum_{n=1}^{\infty} \frac{n\pi}{2}$ converge or diverge? Explain.
37. Explain why, for each n , at least one of $\{|\sin n|, |\sin(n+1)|, \dots, |\sin(n+6)|\}$ is larger than $1/2$. Use this relation to test convergence of $\sum_{n=1}^{\infty} \frac{|\sin n|}{\sqrt{n}}$.
38. Suppose that $a_n \geq 0$ and $b_n \geq 0$ and that $\sum_{n=1}^{\infty} a_n^2$ and $\sum_{n=1}^{\infty} b_n^2$ converge. Prove that $\sum_{n=1}^{\infty} a_n b_n$ converges and $\sum_{n=1}^{\infty} a_n b_n \leq \frac{1}{2} \left(\sum_{n=1}^{\infty} a_n^2 + \sum_{n=1}^{\infty} b_n^2 \right)$.
39. Does $\sum_{n=2}^{\infty} 2^{-\ln \ln n}$ converge? (Hint: Write $2^{\ln \ln n}$ as a power of $\ln n$.)
40. Does $\sum_{n=2}^{\infty} (\ln n)^{-\ln n}$ converge? (Hint: Use $n = e^{\ln n}$ to compare to a p -series.)
41. Does $\sum_{n=2}^{\infty} (\ln n)^{-\ln \ln n}$ converge? (Hint: Compare a_n to $1/n$.)
42. Show that if $a_n \geq 0$ and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges. If $\sum_{n=1}^{\infty} a_n^2$ converges, does $\sum_{n=1}^{\infty} a_n$ necessarily converge?
43. Suppose that $a_n > 0$ for all n and that $\sum_{n=1}^{\infty} a_n$ converges. Suppose that b_n is an arbitrary sequence of zeros and ones. Does $\sum_{n=1}^{\infty} a_n b_n$ necessarily converge?
44. Suppose that $a_n > 0$ for all n and that $\sum_{n=1}^{\infty} a_n$ diverges. Suppose that b_n is an arbitrary sequence of zeros and ones with infinitely many terms equal to one. Does $\sum_{n=1}^{\infty} a_n b_n$ necessarily diverge?
45. Complete the details of the following argument: If $\sum_{n=1}^{\infty} \frac{1}{n}$ converges to a finite sum s , then $\frac{1}{2}s = \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots$ and $s - \frac{1}{2}s = 1 + \frac{1}{3} + \frac{1}{5} + \dots$. Why does this lead to a contradiction?

46. Show that if $a_n \geq 0$ and $\sum_{n=1}^{\infty} a_n^2$ converges, then $\sum_{n=1}^{\infty} \sin^2(a_n)$ converges.
47. Suppose that $a_n/b_n \rightarrow 0$ in the comparison test, where $a_n \geq 0$ and $b_n \geq 0$. Prove that if $\sum b_n$ converges, then $\sum a_n$ converges.
48. Let b_n be an infinite sequence of zeros and ones. What is the largest possible value of $x = \sum_{n=1}^{\infty} \frac{b_n}{2^n}$?
49. Let d_n be an infinite sequence of digits, meaning d_n takes values in $\{0, 1, \dots, 9\}$. What is the largest possible value of $x = \sum_{n=1}^{\infty} \frac{d_n}{10^n}$ that converges?
50. Explain why, if $x > 1/2$, then x cannot be written as $x = \sum_{n=2}^{\infty} \frac{b_n}{2^n}$, where $b_n = 0$ or 1 (and $b_1 = 0$).
51. [T] Evelyn has a perfect balancing scale, an unlimited number of 1 kg weights, and one each of 1/2 kg, 1/4 kg, 1/8 kg, and so on. She wishes to weigh a meteorite of unspecified origin to arbitrary precision. Assuming the scale is big enough, can she do it? What does this have to do with infinite series?
52. [T] Robert wants to know his body mass to arbitrary precision. He has a big balancing scale that works perfectly, an unlimited collection of 1 kg weights, and nine each of 0.1 kg, 0.01 kg, 0.001 kg, and so on. Assuming the scale is big enough, can he do this? What does this have to do with infinite series?
53. The series $\sum_{n=1}^{\infty} \frac{1}{2n}$ is half the harmonic series and hence diverges. It is obtained from the harmonic series by deleting all terms in which n is odd. Let $m > 1$ be fixed. Show, more generally, that deleting all terms $1/n$ where $n = mk$ for some integer k also results in a divergent series.
54. In view of the previous exercise, it may be surprising that a subseries of the harmonic series in which about one in every five terms is deleted might converge. A depleted harmonic series is a series obtained from $\sum_{n=1}^{\infty} \frac{1}{n}$ by removing any term $1/n$ if a given digit, say 9, appears in the decimal expansion of n . Argue that this depleted harmonic series converges by answering the following questions.
- How many whole numbers n have d digits?
 - How many d -digit whole numbers, denoted by $h(d)$, do not contain 9 as one or more of their digits?
 - What is the smallest d -digit number $m(d)$?
 - Explain why the depleted harmonic series is bounded by $\sum_{d=1}^{\infty} \frac{h(d)}{m(d)}$.
 - Show that $\sum_{d=1}^{\infty} \frac{h(d)}{m(d)}$ converges.
55. Suppose that a sequence of numbers $a_n > 0$ has the property that $a_1 = 1$ and $a_{n+1} = \frac{1}{n+1} S_n$, where $S_n = a_1 + \dots + a_n$. Can you determine whether $\sum_{n=1}^{\infty} a_n$ converges? (Hint: S_n is monotone.)
56. Suppose that a sequence of numbers $a_n > 0$ has the property that $a_1 = 1$ and $a_{n+1} = \frac{1}{(n+1)^2} S_n$, where $S_n = a_1 + \dots + a_n$. Can you determine whether $\sum_{n=1}^{\infty} a_n$ converges? (Hint: $S_2 = a_2 + a_1 = a_2 + S_1 = a_2 + 1 = 1 + 1/4 = (1 + 1/4)S_1$, $S_3 = \frac{1}{3^2} S_2 + S_2 = (1 + 1/9)S_2 = (1 + 1/9)(1 + 1/4)S_1$, etc. Look at $\ln(S_n)$, and use $\ln(1+t) \leq t$ for $t > 0$.)